

Image Analysis in Microscopy: Mathematical Morphology and Hough spaces

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Neighborhood Filter Classes

Location center pixel	Input center pixel location	Coordinate transformation
Pixel values involved		
Only intensity values	Rank Filter	
Both intensity and location values	Convolution Filter	Geometrical transformation with Intensity Interpolation
	Morphological Transform	

Mathematical Morphology

J. Serra

Set operations on 2D and 3D shapes:

Shape arithmetic

2D: binary images

Set A (fore- or background pixels)

Structuring element B

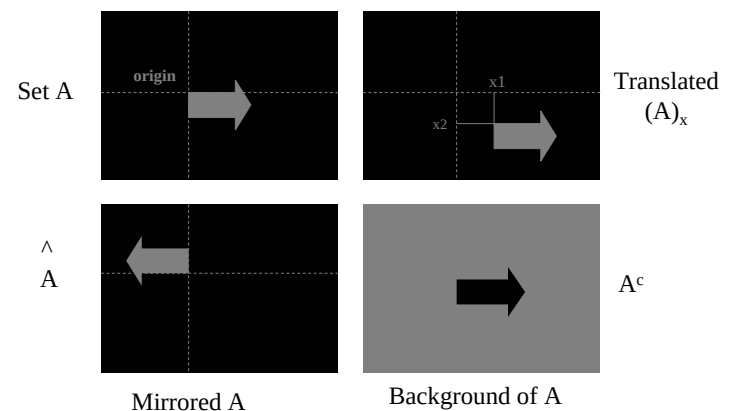
3D: grey-level images

Below surface or above surface

Basic Operations for:

- Noise/detail removal
- contour extraction
- connected areas
- skeletons

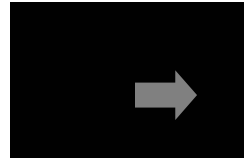
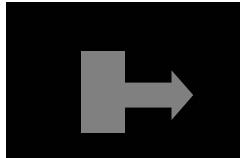
Definitions Math Morph



Set Operations with shape

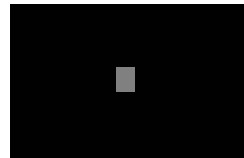
union

$A+B$
or



$A-B$
Only A

$B-A$
Only B

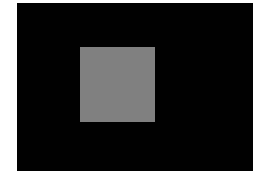


intersection
 $A \cap B$
and

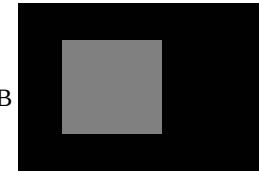
Dilation of A with structuring element B

Origin in the middle of B

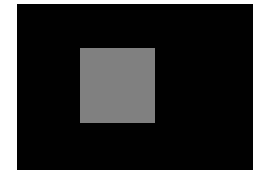
B1



$A \oplus B$



B2



$A \oplus B$

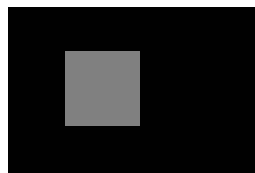


Extent of union when origin of B within A

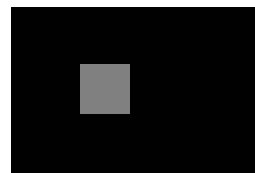
Erosion of A with structuring element B

Origin in the middle of B

B1

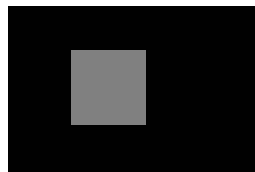


$A \ominus B$

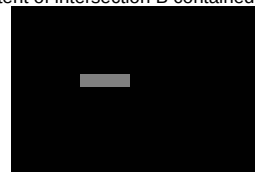


Extent of intersection B contained in A

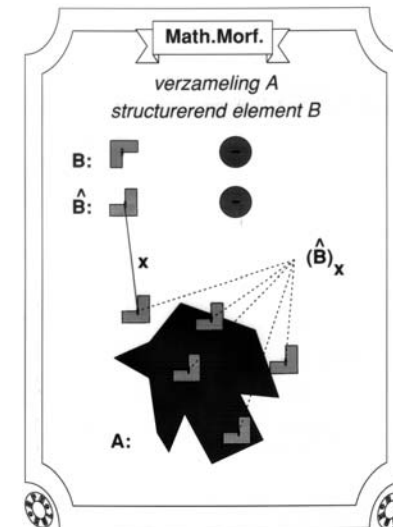
B2

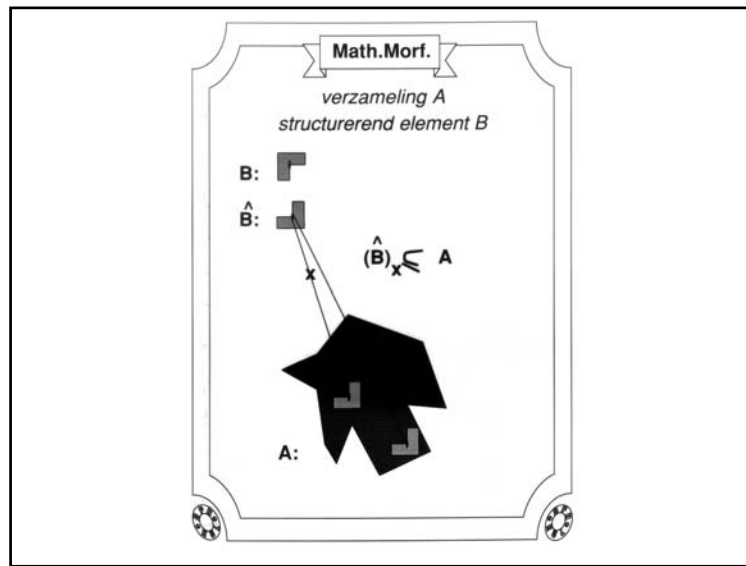
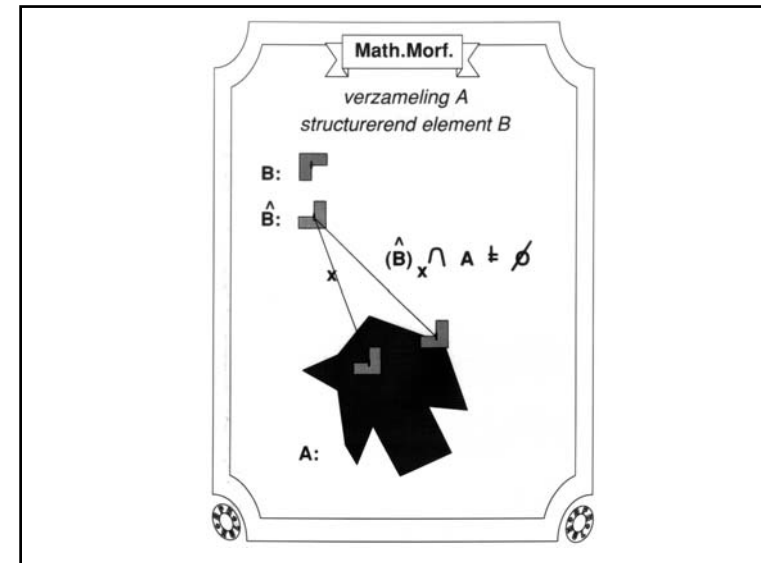
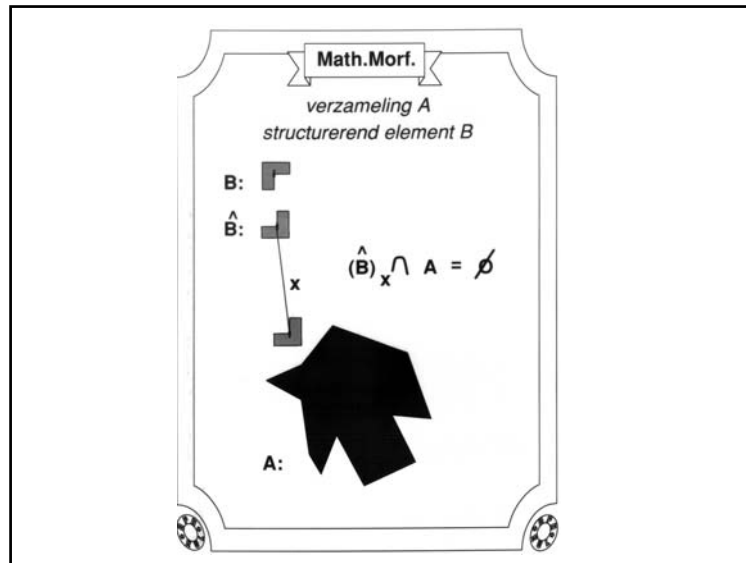


$A \ominus B$



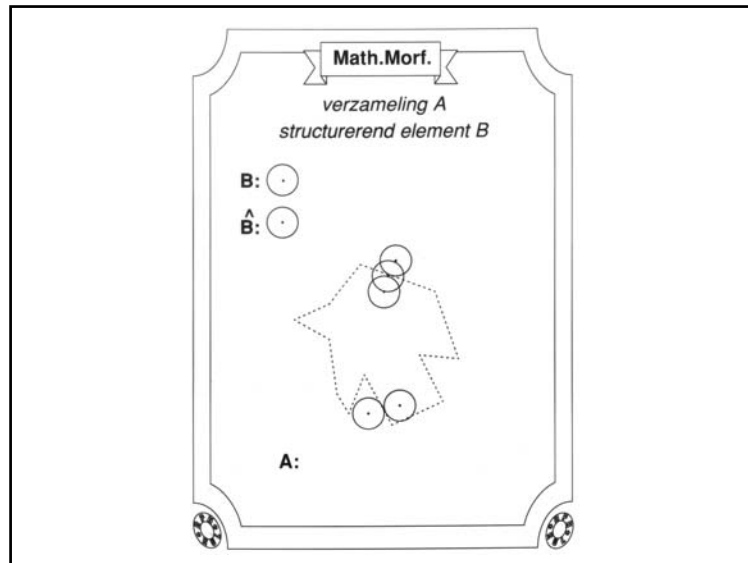
Erosion of black shape A = dilation of background





Contour Probe or Rolling Circle

- Since B completely outside A or B completely inside A doesn't change the resulting shape of A one might as well say that the effect of B is determined by the way its shape fits into the contour of A from either the inside or outside.
- So one might consider structuring element B as a contour probe (usually a rolling circle)

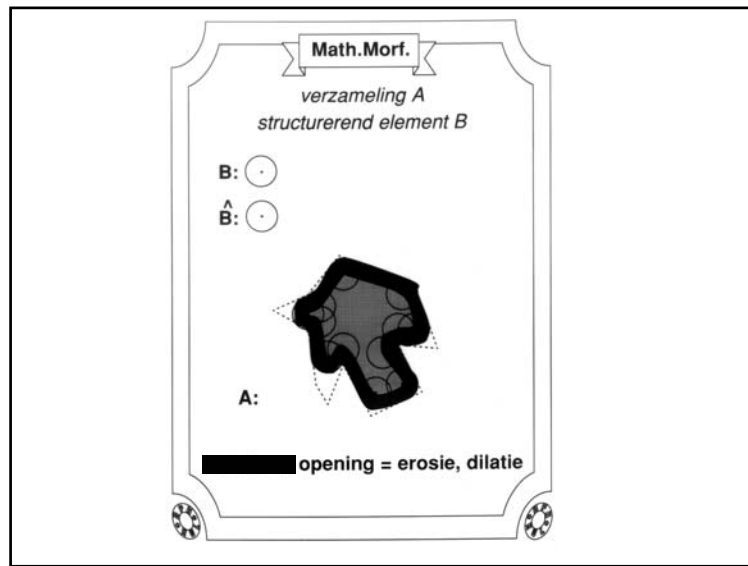


Opening

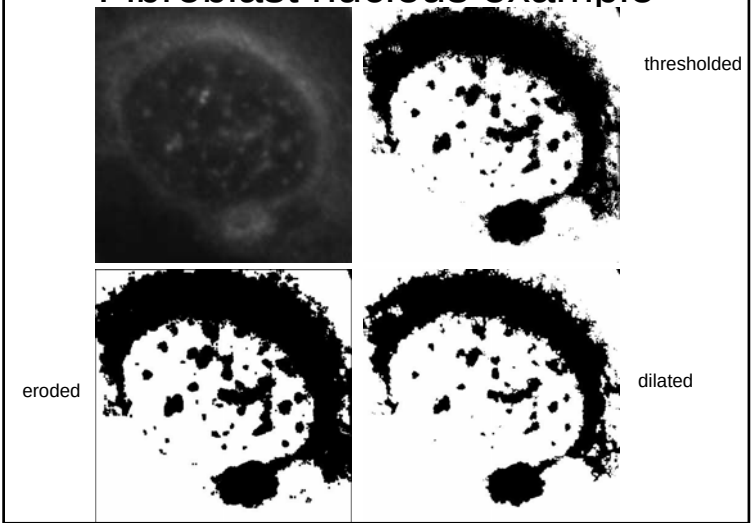
First Erosion then Dilation

$$A \bullet B = (A \ominus B) \oplus B$$

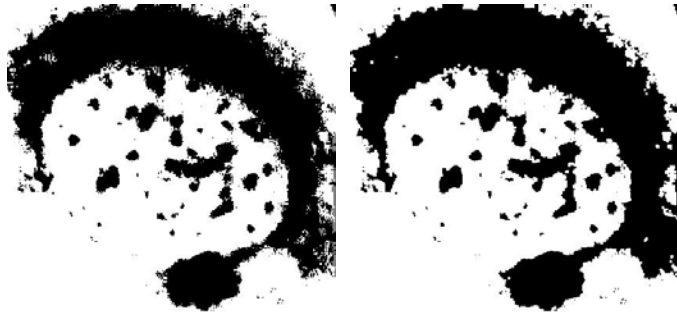
Removes thin connections and thin extensions
And small loose areas



Fibroblast nucleus example



Fibroblast example



Binarized start shape

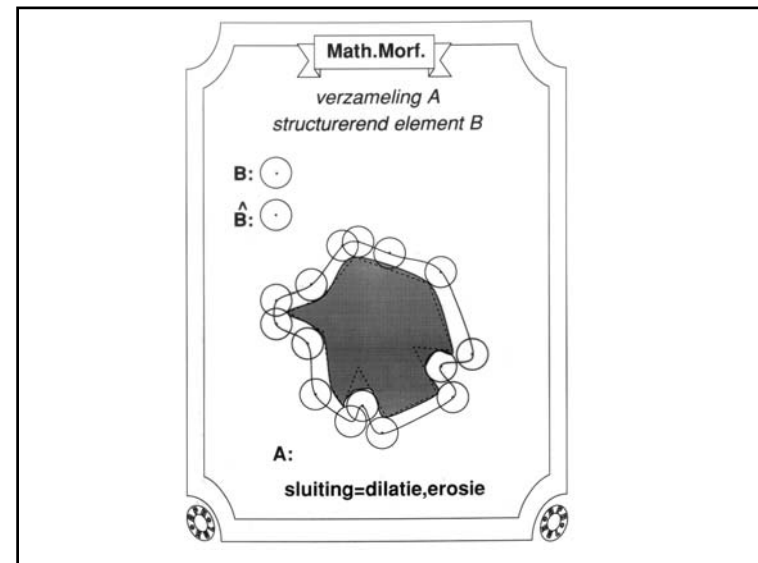
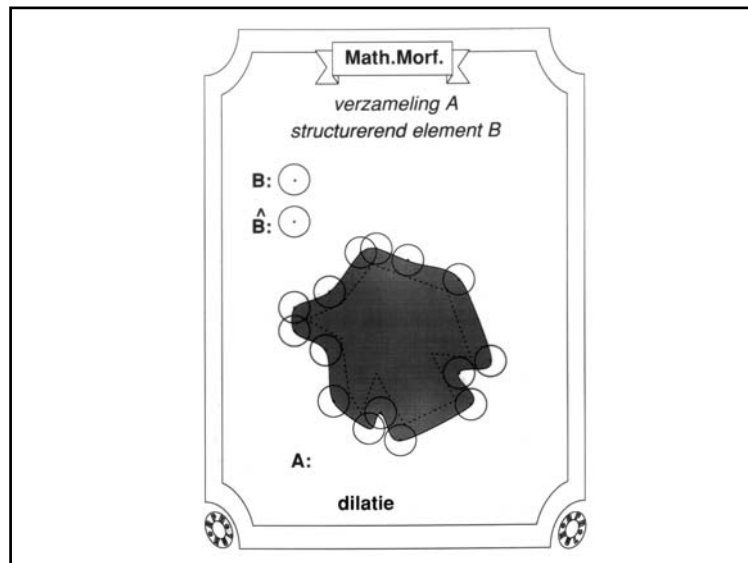
Opened shape
Small white areas removed

Closing

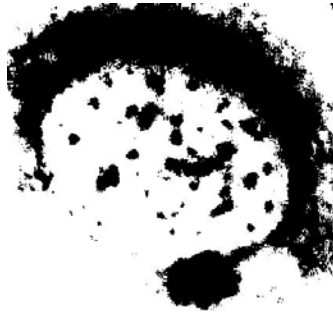
First Dilation then Erosion

$$A \bullet B = (A \oplus B) \ominus B$$

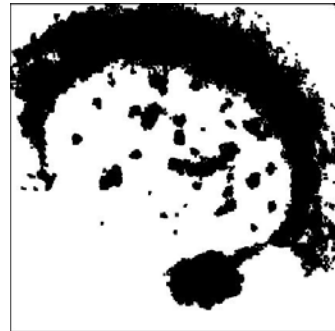
Removes thin intrusions and small holes



Fibroblast example



Binarized start shape



Closed shape
Small black areas removed

Opening and Closing

Morphological Filtering

$$(A \bullet B) \ominus B$$

Removes small detail

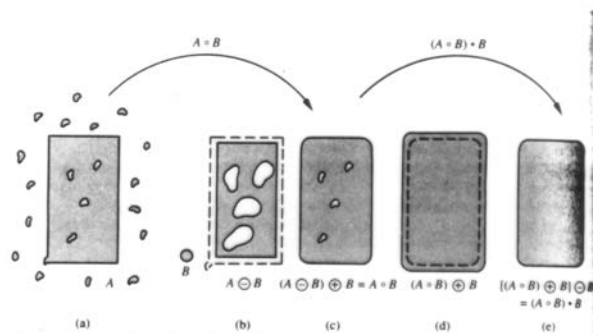


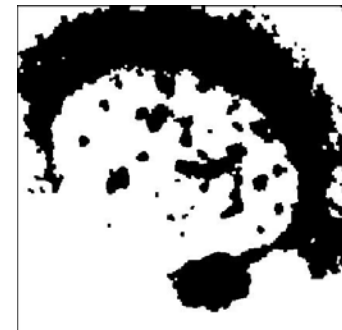
Figure 8.31 Morphological filtering: (a) original, noisy image; (b) result of erosion; (c) opening of A; (d) result of performing dilation on the opening; (e) final result showing the closing of the opening. (Adapted from Giardina and Dougherty [1988].)

Fibroblast example



Binarized start shape

Morphologically cleaned
After opening and closing
Removal of small detail



2D Morphological algorithms

- In a following lecture:
- Contour extraction
- Region filling
- Connected components
- skeletons

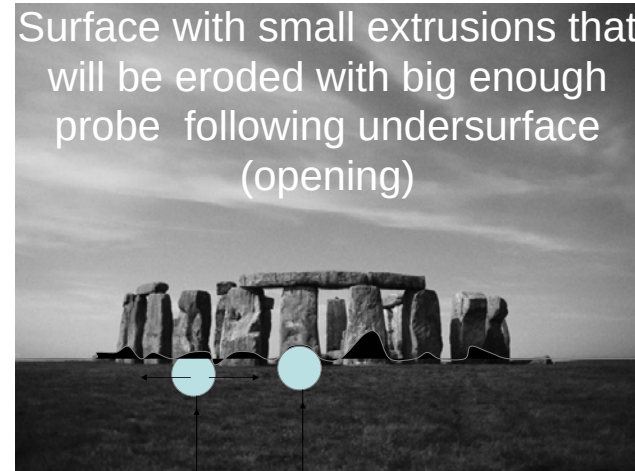
Morphological Transformation of Gray value images

- Consider gray value image as a binary mountain area where the gray value determines the local height and thus the local surface (relief map)
- Consider the structuring element as a small three dimensional probe (usually rolling ball)

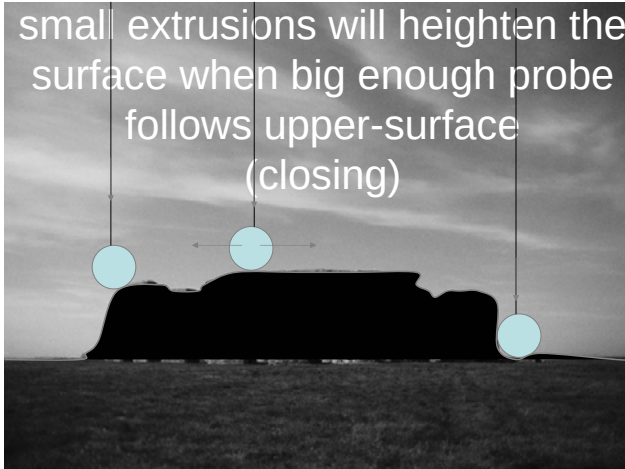
Eroding or dilating a mountainous area is determined by its surface

- Like with 2D binary shapes, in 3D relief maps the result of probe, relief interaction is determined by the result near the surface
- So one might as well speak about surface probing with a probe shape lowered from either the sky or lifted from down under
- Much like how the AFM
- (Atomic Force Microscope) forms a physical relief map by probing a surface at atomic levels with an atomic size tip

Surface with small extrusions that will be eroded with big enough probe following undersurface (opening)



small extrusions will heighten the surface when big enough probe follows upper-surface (closing)



3D Morphological smoothing

- A morphological opening followed by a closing eliminates both small extrusions and intrusions at the gray level surface and
- As a result produces a smoothed image

Comparison with Rank filters

- Min-filters remove salt noise (light outliers)
- Max-filters remove pepper noise (dark outliers)
- Median filter removes salt and pepper noise (all outliers)
- The smoothed result of the median filter comes closest to the morphological smoothing filter

Morphological Gradient

- Morphological gradient: difference of dilated and eroded gray values
- $G = (I \oplus B) - (I \ominus B)$
- Thickness of the area between the dilated and eroded surface



3D morphological Top Hat Transform

- $H = A - (A \circ B)$
- original minus opened image



Second Hour: Special features: parameters Hough spaces

Image transformations: parameter spaces

The original image intensity domain may not be the best choice for problems like:

- Find straight-line segments in specific directions
- Find circles or circular arcs
- Locate regular patterns
- Find object boundaries
- Locate specific surface orientations

Hough transform: condensing (points on) line information

Analytic line: $y = m \cdot x + c$ (slope and cut-off)

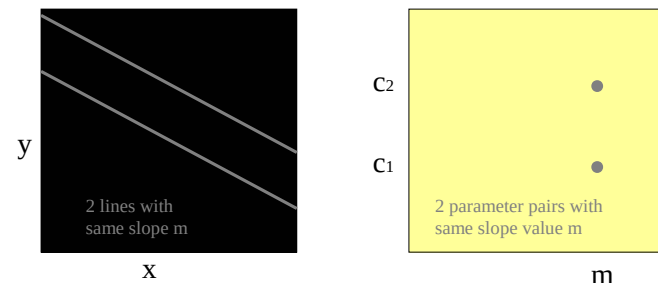
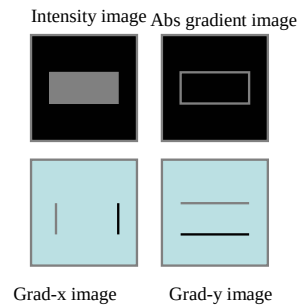


Image space (edges) > hor size Image of Line parameter space
pixels to represent line 1 "pixel" to represent whole line

Hough accumulator scheme $n(m,c)$

- Use local line support in (edge) image to raise corresponding discretized parameter cell (array of accumulators or counters):
- For image point data: raise all accumulator cells that could describe a line through point
- For edge point data: use gradient data to raise specific accumulator that best describes line perpendicular to gradient in that edge point
- Threshold accumulators to find peaks in parameter space (best candidates for line in image/edge space)

Gradient images to support Hough accumulation scheme



- For all x,y with $\text{absgrad} > \text{threshold}$
- - determine corresponding slope m from $-\text{grad-y}/\text{grad-x}$
- - determine cut-off c from $c=y-m*x$
- - raise $n(m,c)$

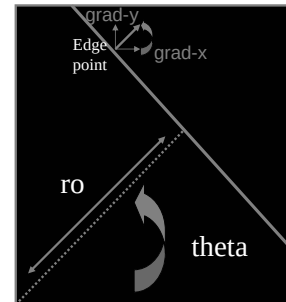
Better Hough spaces

- Original m,c parameter space is unbounded
 - discretization difficult
- Bounded line parameter spaces:
 - (polar coordinate) perpendicular line space:

$$x * \cos(\theta) + y * \sin(\theta) = r_0$$
 - Foot of perpendicular line space:

$$(x - x_l) * x_l + (y - y_l) * y_l = 0$$

Perpendicular Line Space: θ, r_0



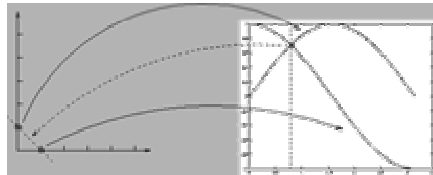
- $0 < r_0 < n \sqrt{2}$
- $-90 < \theta < 180$
- Theta from:

$$\text{tg}(\theta) = \text{grad-y}/\text{grad-x}$$

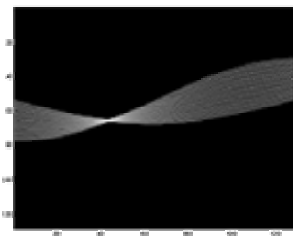
$$\theta = \text{arctg}(\text{grad-y}/\text{grad-x})$$
- Ro from:

$$r_0 = x * \cos(\theta) + y * \sin(\theta)$$

Single Line detection example



Each point with high enough gradient in image space can be addressed by each point on the sinusoid In ro-theta parameter space



The intersection of all sinusoids associated with points on one line can be found by thresholding the parameter space accumulators

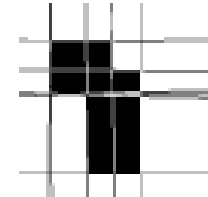
De-Houghing



Input image of overlapping rectangles

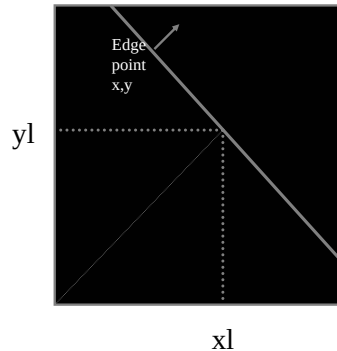


Hough accumulator values



Input image + Reconstructed lines From thresholded Accumulator space

Foot of Perpendicular Line Space: x_l, y_l



$$-n/2 < x_l < n; 0 < y_l < 3n/2$$

x_l/y_l from:

$$y_l/x_l = \text{grad-}y/\text{grad-}x$$

$$w = (x * \text{grad-}x + y * \text{grad-}y) / (\text{grad-}x * \text{grad-}x + \text{grad-}y * \text{grad-}y)$$

$$x_l = \text{grad-}x * w$$

$$y_l = \text{grad-}y * w$$

Accumulator image can be superimposed on (edge) image

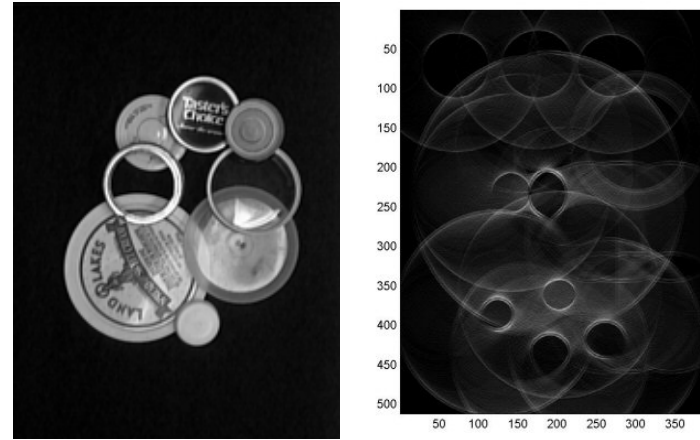
Generalized Hough transform

- The Hough approach can be applied to any parametrized shape description like:
 - circle detection: 3 dim x_m, y_m, r space
 - ellipse detection: 5 dim x_m, y_m, r_a, r_b, ϕ
 - parametrized contour: n dim
- For higher dimensional parameter spaces the accumulator arrays may be prohibitively large and nr of counts per cell very low

Circle parameter space

- 3 dimensional space:
- Focus (x, y coordinates) 2 dimensional space
- Radius (r coordinate) 1 dimensional space
- Results of 3D accumulators shown in 2D:
- Projection on a side

Circle detection example



2D Projection view of a 3D param space